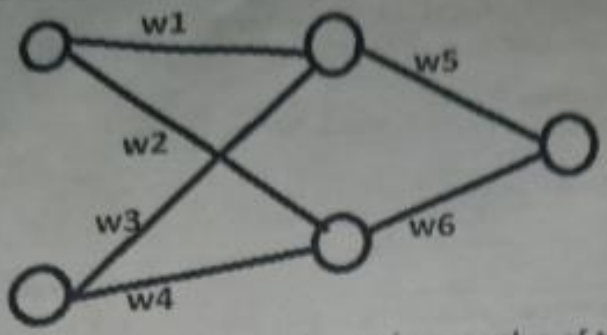
(Please write your Enrollment Number) Enrollment No. 0160117202

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B.Tech CSE-AI, Semester:5th
(November-December, 2023)

Subject Code: BAI 305	Subject: Deep Learning - I
Time :3 Hours	Maximum Marks :60

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(5*4=20)	
	(a) Consider a two-input neuron whose weights (w1, w2) are initialized as (0.5, 0.8) with objective function (f(x1, x2) = 3x12+5x2+2). What are the values of (w1, w2) after each iteration of the perceptron learning algorithm for two iterations.		
	(b) Why are activation functions necessary in the hidden layers of ANNs? Is it beneficial to apply different activation functions in different layers of a neural network? Justify		
	(c) Explain the differences between ReLU, Sigmoid, and Tanh. In what scenarios might one be preferred over the other?		
	(d) Is initializing all weights of a CNN model as zero a wise decision? Why or why not? Would incorporating Batch Normalization into the same CNN model be beneficial for CNN models? Justify.		
	UNIT-I		
Q2	a) Discuss the MP-Neuron unit along with the thresholding function. b) "Perceptron is universally applicable to all real-world applications". Validate the statement with a suitable example. c) Discuss the perceptron learning algorithm in detail. What is the reason behind two separate weight updation equations for positive and negative samples?	(3+2+5=10)	
Q3	a) For a neuron that accepts five real values as input and weights are defined as (w1, w2, w3, w4, w5) = (1.0, 0.5, 0.2, 0.4, 0.9) with theta as 0.5. For following inputs, predict the output of this neuron: (i) x1:0.9,x2:0.1,x3:0.49,x4:0.63,x5:0.33. (ii) x1:0.2,x2:0.4,x3:0.91,x4:0.38,x5:0.20. b) What are the advantages of the sigmoid function over the thresholding function? c) What are the threshold values in MP-Neuron for implementing NOR, AND, and NOT Boolean functions? Provide a justification for the chosen threshold values.	(3+2+5=10)	
	UNIT-II		
Q4	Consider the given neural network with single input as [4, 5] and output as [7]. The initial weight values (w1, w2, w3, w4, w5, and w6) are [0.2, 0.2, 0.3, 0.1, 0.4, 0.2] and the loss function is mean squared error with learning rate as 0.01. Assuming there are no activation function and biases, answer the following with 2 decimal places:	(10)	

	 a) Find the prediction value and error value of the neural network in the first forward pass. b) Find the updated values of weights (w1 to w6) after backpropagation algorithm once.		
Q5	a) Differentiate between L1 and L2 regularization techniques. b) Is dropout layer active during the testing of a deep neural network? Justify c) Explain different data augmentation techniques. How is it useful for regularization?	(2.5+2.5+5 =10)	
	UNIT-III		
Q6	a) Discuss the architecture of Long Short-Term Memory (LSTM) networks. How does it differ from standard RNNs. b) Why does RNN perform on natural language processing, time series analysis, and other sequential data tasks? c) What are the roles of hidden states and recurrent connections in an RNN?	(5+2.5+2.5 =10)	
Q7	a) Explain the architecture of Gated Recurrent Units (GRUs), and how do they differ from LSTM units? b) What are Recurrent Neural Networks (RNNs)? What are the different application areas of RNN and how do they differ from feedforward neural networks?	(5+5=10)	
	UNIT-IV		
Q8	Consider a convolution neural network consisting of two convolutional layers and one max-pooling layer. The shape of any matrix is in the form of channel x height x width. The size of the input image is 3 x 224 x 224. The first convolutional layer has following configuration: Kernel size: 5x5, No. of kernels: 96, Stride: 1, Padding: 1. The second convolutional layer has following configuration: Kernel size: 6x6, No. of kernels: 128, Stride: 2, Padding: 1. The maxpooling layer has following configuration: Kernel size: 2x2, Stride: 2. Based on the given architecture, answer the following questions: a) What is the size of the output feature maps after first convolutional layer, second convolutional layer and maxpooling layer? b) What is the depth of each convolutional kernel in first and second convolutional layers?	(10)	
Q9	a) Explain the concept of Transfer learning with an example? b) Differentiate between AlexNet and VGG? c) Discuss the architecture of InceptionNet. How does the Inception module contribute in improving the performance of InceptionNet?	(2.5+2.5+5 =10)	